

Alexander T.J. Barron

✉ cogentmentat@gmail.com | 🏠 cogentmentat.github.io/academic | 🐙 github.com/CogentMentat | 🔗 linkedin.com/in/alexander-tj-barron

Summary

Ph.D. with years of experience tackling new and unstructured data sets to solve novel problems, focused on data acquisition, NLP, and machine learning.

- Extensive experience in data collection and analysis at terabyte-scale, from design to implementation to results.
- Broad experience with machine learning, deep learning, and nonparametric statistics.
- Excellent programming skills in Python and the Jupyter/Pandas stack. Now force-multiplied with LLMs, coding assistants, and the cloud.
- Practiced experience working with a diverse team, translating abstract questions into measurable analysis.

Skills

Programming/compute	Python (pandas, jupyter, scipy, numpy, matplotlib, scikit-learn, pytorch, tensorflow, etc.), R, SQL, Bash, mongodb, Entire AWS stack, Google Cloud Platform, coding assistant management, agents, HPC
Algorithms/techniques	NLP, ML, data mining, sentiment analysis, nonparametric statistics, bootstrapping, regression, topic modeling, clustering, embeddings, LLMs, fine-tuning, contrastive learning, Named Entity Recognition, prompting
Communication	Writer of 5+ publications; Associate instructor for multiple courses; Multiple conference presentations
Mentorship	Management of 2 undergraduates' research work through writing and publication
Miscellaneous	Git, Github, *nix systems, SLURM, AWS, GCP, Docker, Gephi, $\text{\LaTeX}$, jazz drumset

Work Experience and Projects

New York University Abu Dhabi & Max Planck Institute for Empirical Aesthetics

Abu Dhabi, UAE & Frankfurt, Germany

Post-Doctoral Associate

September 2023 - present

- Designed and implemented massive audio data ingestion, consisting of global radio and amateur music production. Coupled metadata collection and audio sampling feed continually into munging and analysis pipelines. Iterated application of various models for segmentation, transcription, audio source separation, and Music Information Retrieval tools to measure culturally-relevant signatures in music.

IU School of Informatics, Computing, and Engineering

Graduate Researcher

2014 - 2022

- Measured individual and group power dynamics in political legislative speeches and government formation.
- Revealed differences in network clustering structure in collective identities online, from terabytes of Twitter profile data.
- Analyzed Twitter and Reddit content, part of the DARPA Next Generation Social Science grant, studying identity and group formation.
- Parsed the Bitcoin block chain and developed an indicator of anonymity-conscious decision-making by entities using Bitcoin.

Associate Instructor

2012 - 2021

- Multiple courses: *Information Infrastructure*, *Research Methods: Large-scale Social Phenomena*, *Mathematical Foundations of Informatics*. Topics include code management, security, ethics, Python and unix tools, information theory, social information, introductory game theory, probability, foundational discrete mathematics

Selected academic contributions

- **Alexander T. J. Barron**, Jenny Huang, Rebecca L. Spang, Simon DeDeo. (2018). [Individuals, institutions, and innovation in the debates of the French Revolution](#). *Proceedings of the National Academy of Sciences*, 115 (18).
 - Winner of the 2018 [Cozzarelli prize](#), covered by [Christian Science Monitor](#), [Ars Technica](#), and [MIT Technology Review](#).
- Johan Bollen, Marijn Ten Thij, Fritz Breithaupt, **Alexander T. J. Barron**, Lauren A. Rutter, Lorenzo Lorenz-Luaces, Marten Scheffer. (2021). [Historical language records reveal a surge of cognitive distortions in recent decades](#). *Proceedings of the National Academy of Sciences*, 118 (30).
 - Featured in the [United Nations Human Development Report 2021/2022](#), p. 31.
- **Alexander T. J. Barron**, Johan Bollen. [Quantifying collective identity online from self-defining hashtags](#). (2022). *Scientific Reports*, 12, 15044.

Reviewer: Cognitive Science, PLOS One, ICWSM

Education

Ph.D. in Informatics Complex Systems Track

Indiana University, August 2012 - May 2022

B.S. in Physics and Applied Mathematics *summa cum laude*

University of New Mexico, August 2006 - July 2011

Selected Awards

- 2018 **Cozzarelli Prize in Behavioral and Social Sciences**, one of six such yearly prizes reflecting “scientific excellence and originality” from the National Academy of Sciences, USA.
- 2019-20 **NSF Research Traineeship award**, summer affiliate in Complex Networks and Systems. Total amount: \$10,000.